

a

Controlled evaluation pipeline

Policy input: visual observations · instruction · robot state Policy output: action chunk

PC / WSL

LIBERO Spatial

tasks · scenes
success checks

MuJoCo 3.3.2

physics · EGL
camera rendering

PC-local evaluation

SmoIVLA

PC-local

Jetson Orin Nano

Policy service

SmoIVLA
remote inference

HTTP

Fixed evaluation controls

checkpoint · task suite · MuJoCo/EGL · episode protocol · evaluator

b

Layered policy optimisation

Fixed SmoIVLA checkpoint

no retraining or architecture changes

Software-level optimisation

num_steps · n_action_steps · chunk_size

Scope-aware quantisation

INT8 · INT4 · mixed precision

Hardware acceleration

TensorRT · CUDA Graphs

Incremental end-to-end ablation